

Predicting DNA binding of new
DNA-binding proteins

Project in
Deep Learning in
Computational Biology

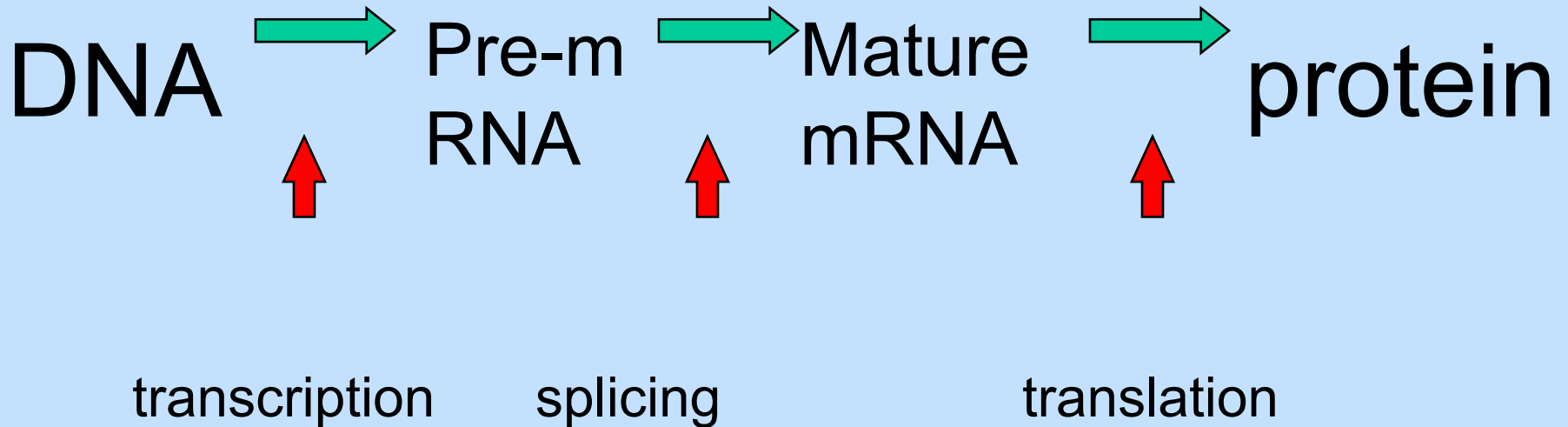
Outline

1. Some biological background again...
2. The project

1. Background

Slides with Ron Shamir and Chaim Linhart,
Computational Genomics TAU

Gene: from DNA to protein



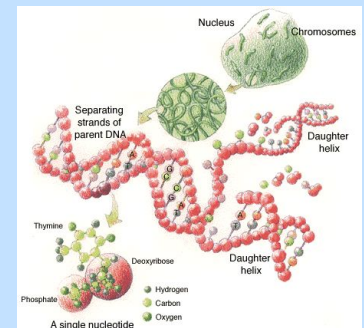
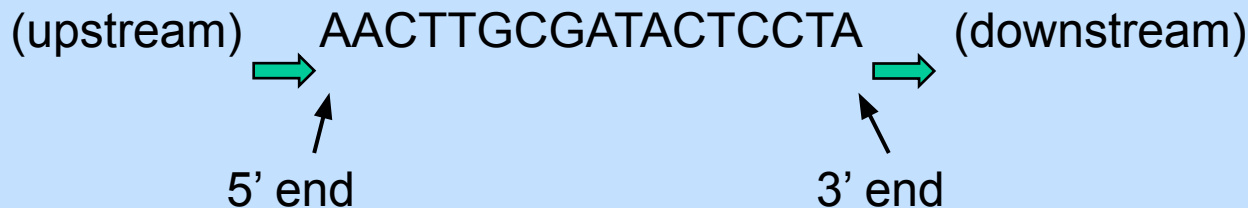
DNA

- DNA: a “string” over the alphabet of 4 **bases** (nucleotides): **{ A, C, G, T }**
- Resides in chromosomes
- Complementary **strands**: A-T ; C-G

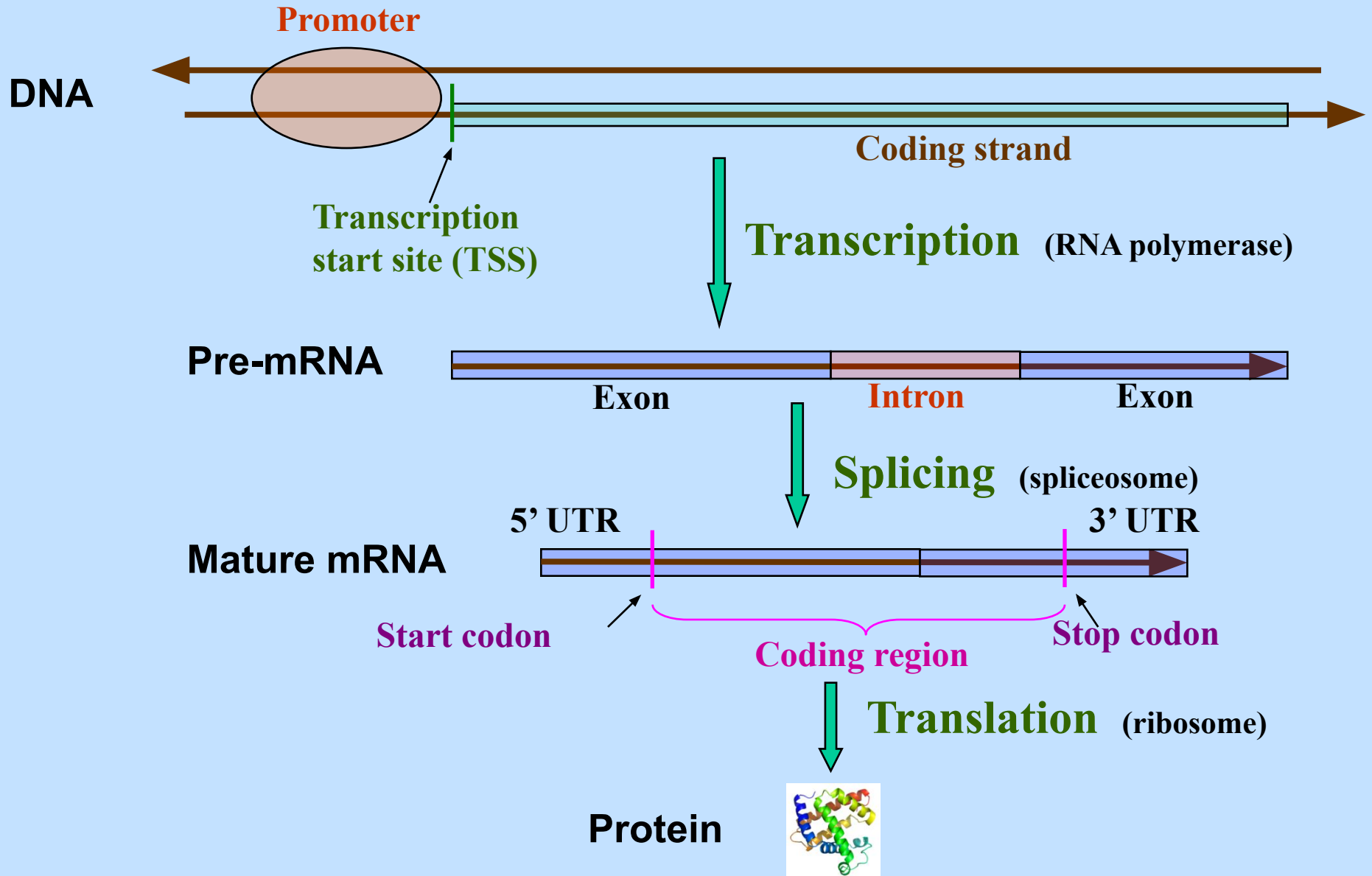
Forward/sense strand: AACTTGCG

Reverse-complement/anti-sense strand: TTGAACGC | |

- Directional: from 5' to 3':



Gene structure (eukaryotes)

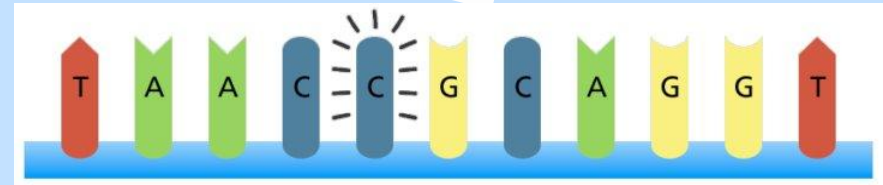
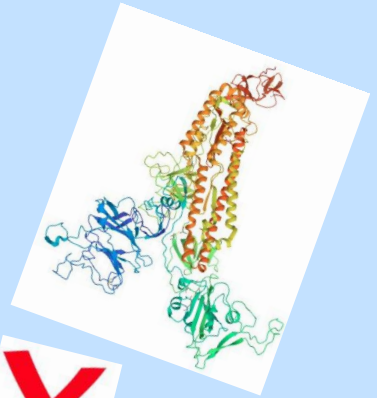
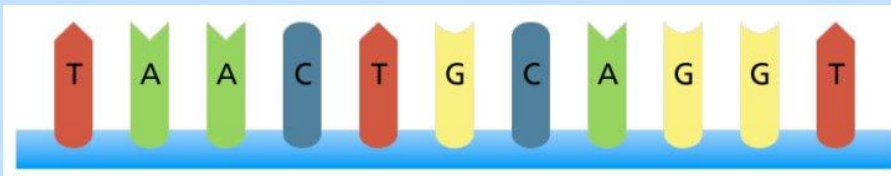
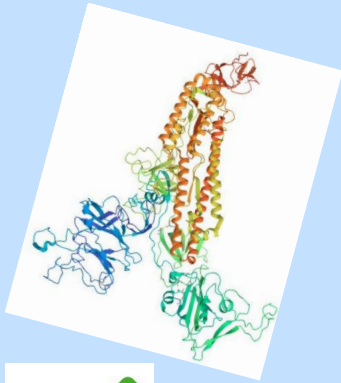


Regulation of Expression

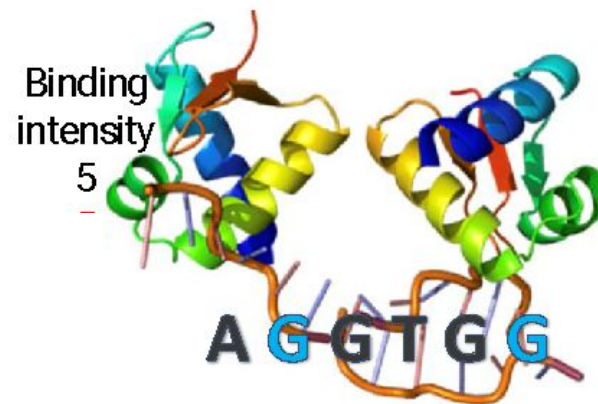
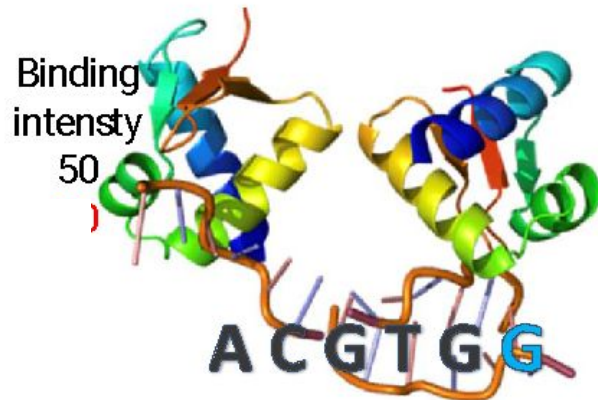
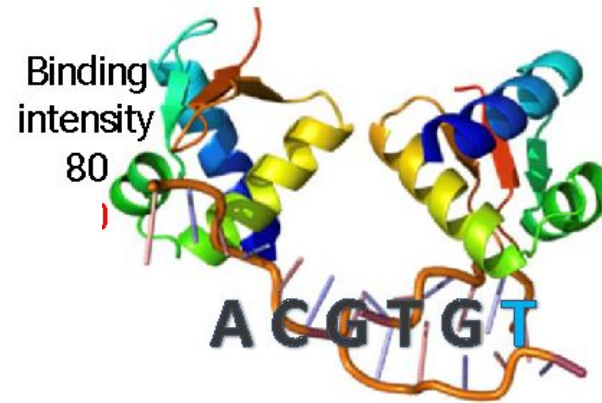
- Each **cell** contains an identical copy of the whole genome - but utilizes only a subset of the genes to perform diverse, unique tasks
- Most genes are highly regulated - their **expression** is limited to specific tissues, developmental stages, physiological condition
- Main regulatory mechanisms:
 - **transcriptional regulation**
 - post-transcriptional regulation

Transcriptional regulation

- DNA-binding proteins (DBPs) are primary regulators of gene expression.
- DBPs recognize **specific DNA binding sites**.



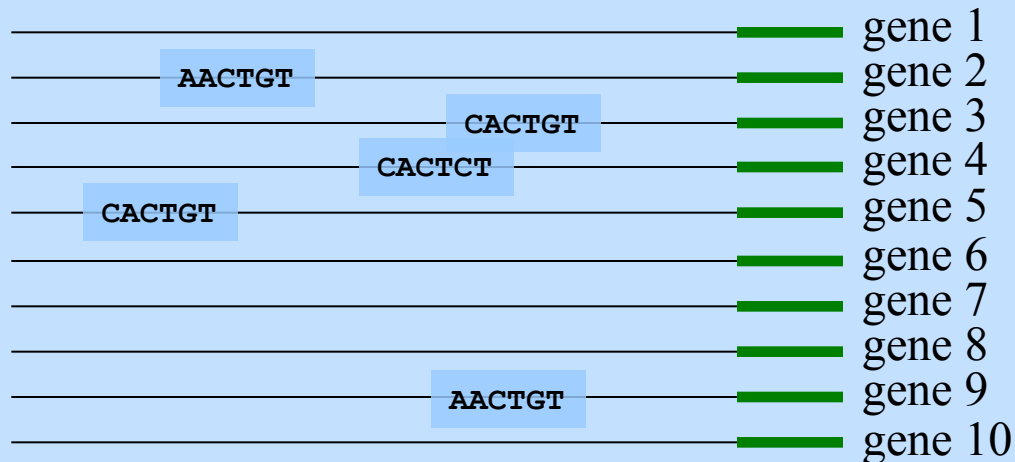
Protein DNA-binding



DBP BS motif models - strings

- Consensus (“degenerate”) string:

$\begin{matrix} A & & C \\ C & A C T & G \end{matrix} T$



How can we use NN for these models?

DBP BS models - PWM

- Position weight matrix (PWM): each position has weights for the 4 possible letters (A, C, G, T)

- For example:

How can we use NN
for this model?

6	5	4	3	2	1	
0	0.2	0.7	0	0.8	0.1	A
0.6	0.4	0.1	0.5	0.1	0	C
0.1	0.4	0.1	0.5	0	0	G
0.3	0	0.1	0	0.1	0.9	T

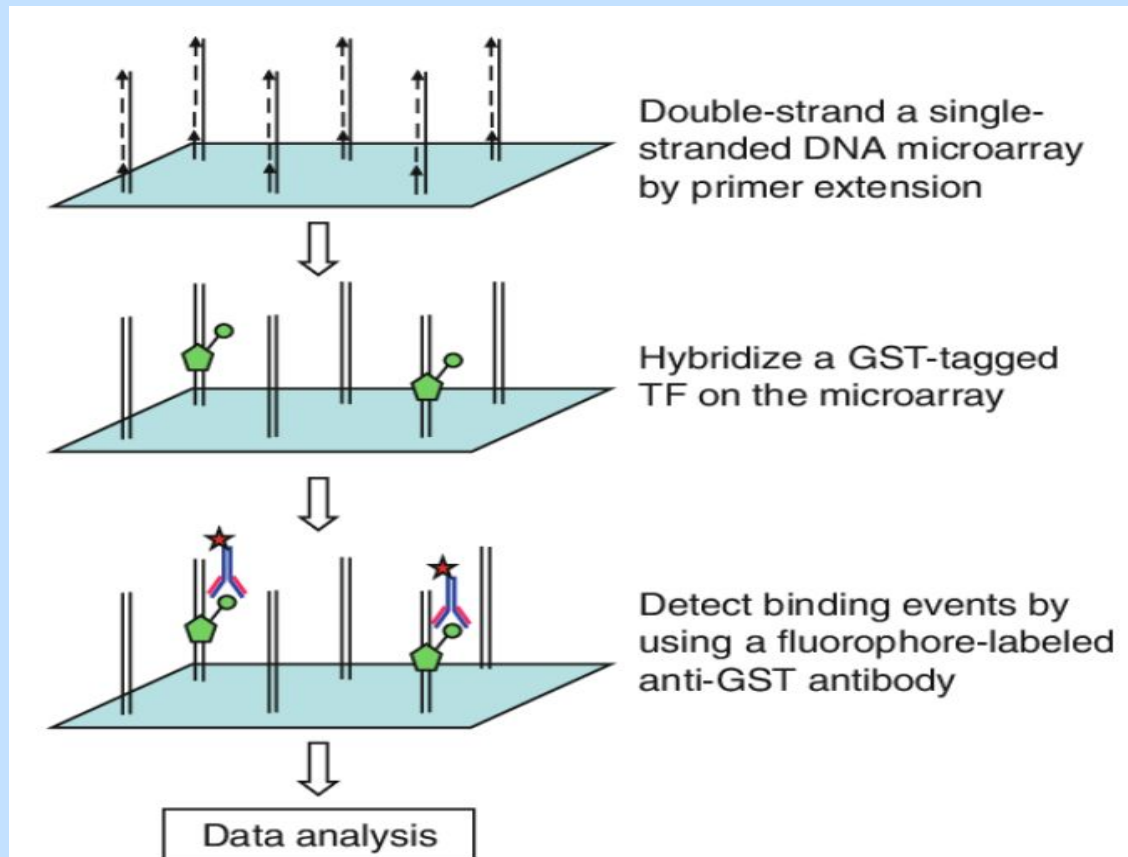
- Logo format:



PBM

Berger, M. F. et al. (2006)

- Generate an DNA pool covering all possible 10-mers, each at least 1 time.



PBM- implementation

- 36nt variable region
- A sequence library covering of all 8-mers, each at least 32 times
- 41,728 probes

The computational challenge

- Input:
 - Amino-acid sequences of new proteins (1 sequence file)
 - a list of DNA sequences (1 sequence file)
- Goal: Predict DNA binding intensity for each DNA sequence
- Intuition: learning a binding model based on a protein's amino acid sequence

The project

General goals

- Research
 - Learn about known solutions
 - Trial and error with training data
- Develop software from A-Z:
 - Design of deep neural networks
 - Implementation
 - Execution & analysis of test data
- A taste of bioinformatics
- Have fun
- Get credit...

The computational task

- Given a set of PBM data of different DBPs
- Learn a model that for a given DBP can score PBM probes
- Main challenges:
 - Prediction performance
 - Prediction runtime

PBP Input

- 1 amino-acid sequence file

<sequence>\n

```
MSSKIFVGGNNVYIGYDNQKALGLILRRFETRKGFGFVTFDDQSRGKLIPYEVVGEVVKLG YRDQR  
G
```

PBM Input

A file with ~41K lines, each containing an DNA sequence of length 36.

<sequence>\n

```
GTAATATTACATTCCGTTTAACTGCGCGCCTACGG
TTTCTCTATGCAGATTAGCTCAATTCCCAATACCTA
GTAATATTCATACGGCCTCGACTCATTACACGGCTT
CTATATGTGCATATATTACCCTACAGGTCGGATAGC
TTCGGATTGGGTGCATAATGAAGCAGGACTATTAAA
CCGAGGTATTTGCATTTTCAGTCGTCTTCTAATGAAG
CCCAGCCGGATTTCATACACGGACCGAGCAATGG
GAAGGATGGTATGCGTTTATTTGCATAGACAACATC
GTTACTATAATTAGCAGCGCGTTTCGGGCCGTTCCAC
...
...
...
ACGTTTGCATTAATTTCCCCACATTATTATGCGGA
CCCAGTTAATTATGCGCCACGAGTCCGTCATAGTTA
ACCTCATTAAATGTATTAAGGTTGTTTAAAGTTGGG
```

- The input sequence file is sorted lexicographically
- The output is a file with a binding intensity for input sequence (in the same order)

Ttrain Test Split

- Train:

- 400 proteins. (training_DBPs.txt)
- 30,000 DNA probes. (training_seqs.txt)
- Binding intensities. (training_data.zip)

Test:

- 64 proteins. (test_DBPs.txt)
- 11,728 DNA probes (test_seqs.txt)

11728

There is no overlap between the protein and DNA sequences used for training and testing

The goal

For each protein in the test set, the model predicts the binding intensity score for every DNA probe sequence in the test set.

For this list we can compare to real values by **Pearson correlation**:

$$\text{cov}(x,y)/(\text{std}(x)*\text{std}(y))$$

Solution example

- Train a model that receives a pair
(amino acid DBP sequence, DNA probe sequence)

And outputs a binding intensity

- Protein representation can be using protein LLMs
- DNA representation by one-hot

Baseline Model (For Performance Comparison)

To evaluate your model, we provide a **baseline results file** (**baseline_results.txt**).

How the baseline was built:

Training data was split into:

- First **380 proteins**
- First **27,000 DNA sequences**

The remaining protein–DNA pairs from the training set were used for evaluation.

Baseline Model

Baseline Prediction Method

- For each test protein-DNA pair:
- **Protein matching:**
 - Find the most similar protein in the training set
 - Similarity is computed using **sequence alignment**
- **DNA matching:**
 - Find the most similar DNA sequence in the training set
 - Similarity is computed using **3-mer count similarity**
- **Prediction:**
- The binding score for the test pair is taken from the **most similar training pair**

Evaluation Metric

- The results in `baseline_results.txt` are **Pearson correlation scores**
- A score is computed **for each protein** separately

Submission

- Electronic design document
- Electronic code submission
- 64 binding intensities files, e.g. DBP1.txt
- Executable / script for running time test

Design document

- 3-5 pages (pdf), Hebrew/English
- Briefly describe main goal, input and output of program
- Describe neural net architecture, algorithms, and scores

References

- Berger, Michael F., et al. "Compact, universal DNA microarrays to comprehensively determine transcription-factor binding site specificities."
- Nature biotechnology 24.11 (2006): 1429-1435.
- ProbeRating: a recommender system to infer binding profiles for nucleic acid-binding proteins